ELECTIVE-5.1

URA412 OPTIMIZATION IN ENGINEERING DESIGN

L	T	P	Cr
3	0	0	3.0

Course Objective: The main objective of this course is to provide the detailed classification of optimization techniques available in order to address wide range of optimization problems. The course will also highlight different solution strategies and performance criterion for applied optimization problems. Through this course, the students will learn how to formulate an engineering optimization problem. The course will also introduce the basics of evolutionary optimization techniques as compared to classical optimization techniques.

Syllabus

Introduction to Optimization: Statement of an Optimization Problem, Classification of Optimization Problems, Optimization Techniques, Solution of Optimization Problems Using MATLAB.

One-dimensional Optimization Methods: Optimality Criteria – necessary and sufficient conditions, Bracketing methods, Region-elimination methods, Point estimation method, Gradient based methods, Sensitivity analysis.

Multi-dimensional Optimization Methods: Optimality Criteria, Unidirectional search, Direct search methods, Gradient-based methods. Conjugate-direction methods, Quasi-Newton methods.

Constrained Optimization Methods: Constrained Optimization Criteria, Penalty Methods, Method of Multipliers, Direct search methods, Linearization methods, Feasible Direction method.

Structured Problems and Algorithms: Integer Programming, Quadratic Programming.

Specialized Optimization Techniques: Introduction to Multi-Objective optimization, Ant Colony Optimization, Particle swarm Optimization.

Course Learning Objectives (CLO)

The students will be able to:

1. solve one-dimensional and multi-dimensional engineering optimization problems.
2. formulate as well as analyze unconstraint and constraint optimization problems.
3. solve special design problems with discrete solutions using Integer programming.

Text Books

1. Deb, K., Optimization for Engineering Design Algorithms and Examples, Prentice Hall of India Pvt. Ltd., (2005), Eighth Print.
2. Deb, K., Multi-objective Optimization using Evolutionary Algorithms, John Wiley and Sons, (2009), First Edition.
3. Rao, S.S., Engineering Optimization Theory and Practice, John Wiley and Sons, (2009), Fourth Edition.

Reference Books

1. Ravindran, A., Ragsdell, K.M., Reklaitis, G.V., Engineering Optimization: Methods and Applications, John Wiley and Sons, (2006), Second Edition.
2. Belegundu, A.D., Chandrupatla, T.R., Optimization Concepts and Applications in Engineering, Cambridge University Press, (2011), Second Edition.
3. Dasgupta, B., Applied Mathematical Methods, Pearson Education India, (2006), First Edition.

Approved in 113th meeting of the Senate held on September 7, 2024, respectively. Further, revised in 1st meeting of the Academic Council held on September 11, 2025.

Evaluation Scheme:

Sl. No.	Evaluation Elements	Weightage (%)
1	MST	30
2	EST	45
3	Sessional (Assignments, Tutorials, Quizzes, Projects, Tests)	25

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